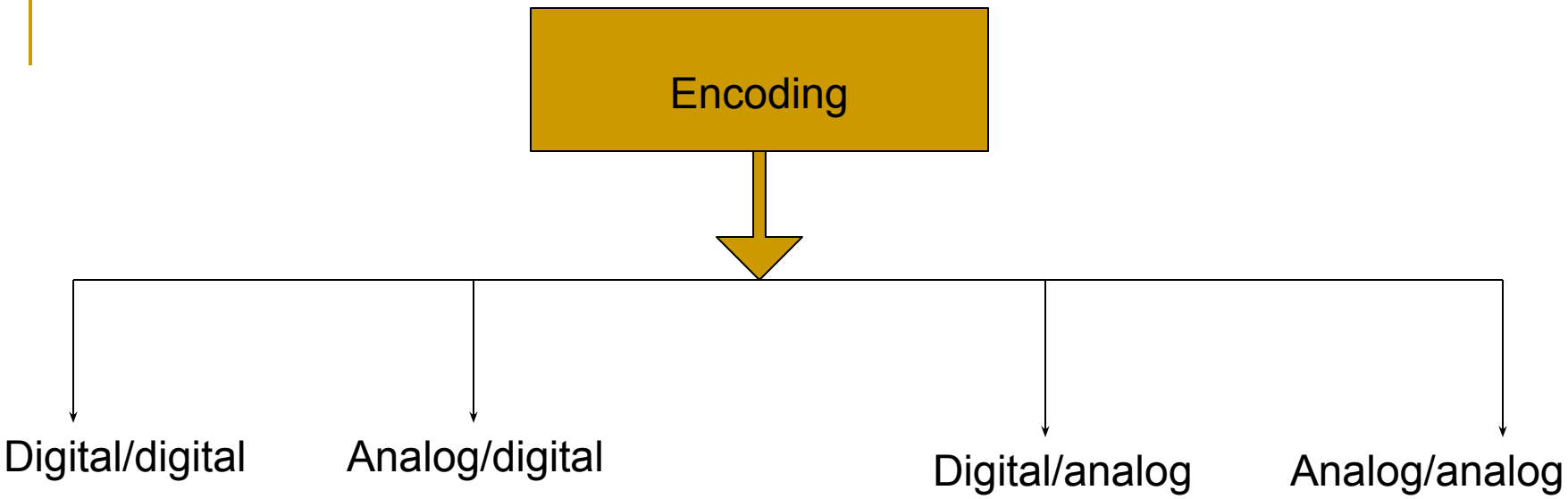
ENCODING

- For the information to be transmitted across a communication medium, it must be encoded into signals.

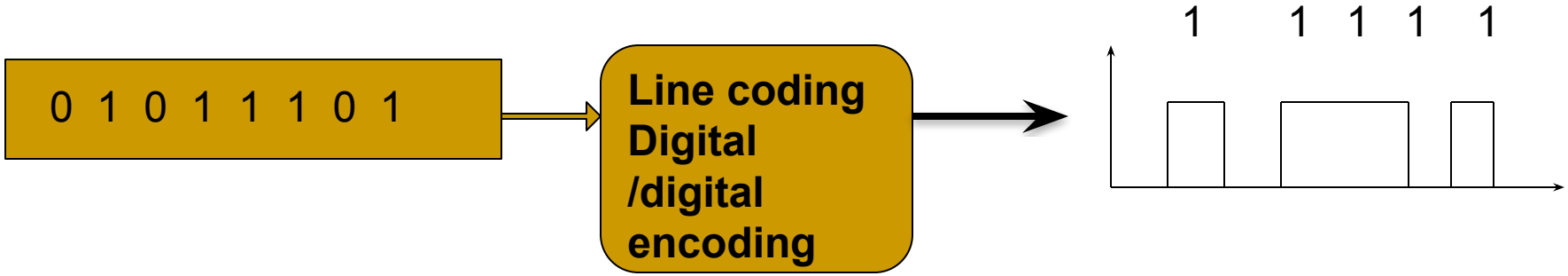
- The signal must be manipulated so that it contains the information that is recognizable to the sender and receiver.

- This process is known as Encoding.


Encoding
Digital/digital
Analog/digital
Digital/analog
Analog/analog

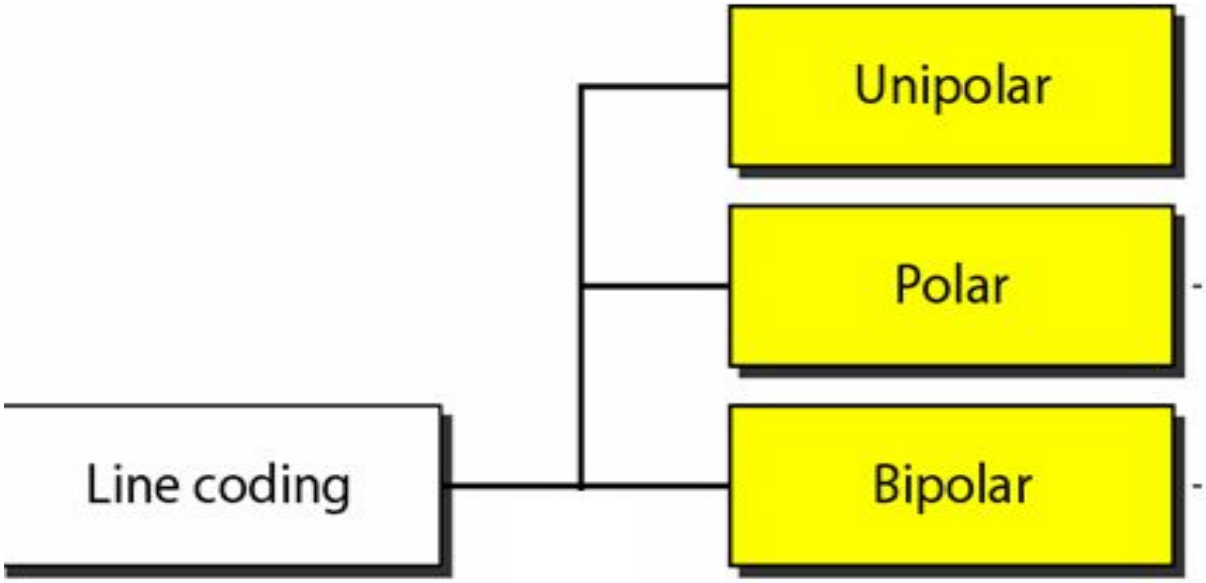
Digital to digital encoding / Line coding

- Line coding is a process of converting digital data to digital signal.
- Data stored in a computer are in the form of 0's and 1's.
- To be carried from one place to another, data must be converted to digital signals.
- Thus line coding converts this sequence of bits(0's and 1's) to a digital signals.
- This is also called Encoding.
- Example:- When we transmit data from computer to printer, both the original data and the transmitted data are digital.

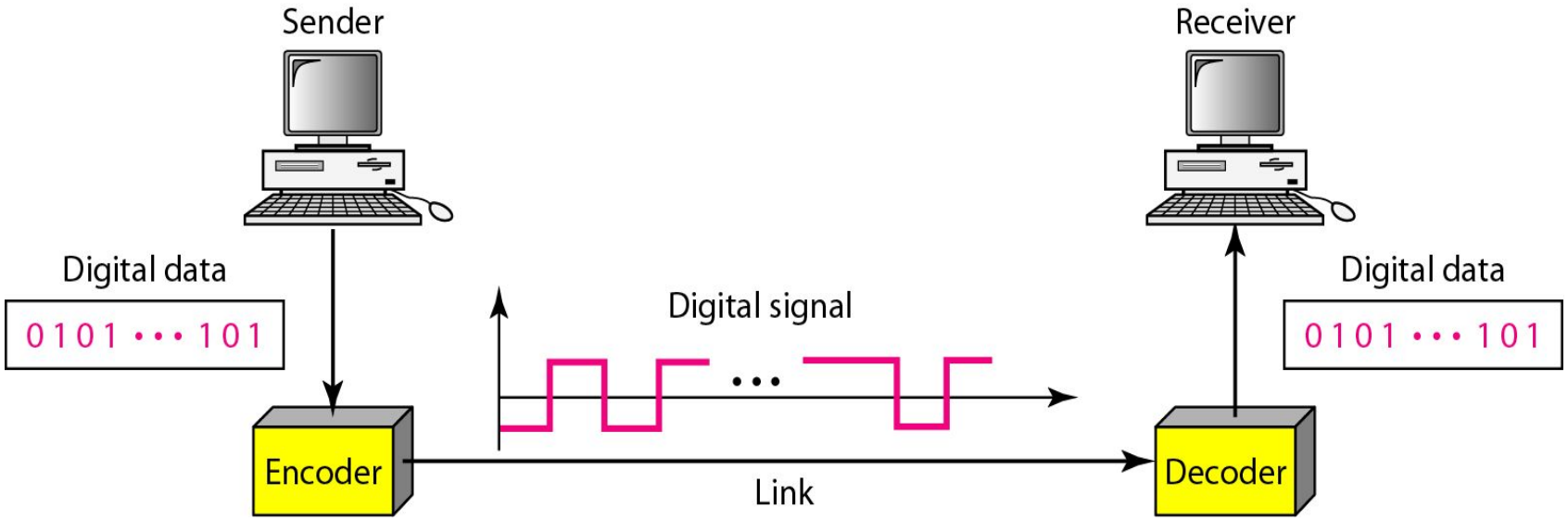


Digital to digital encoding

Line Coding Schemes or 3 types of digital encoding are :-



Line Coding & Decoding

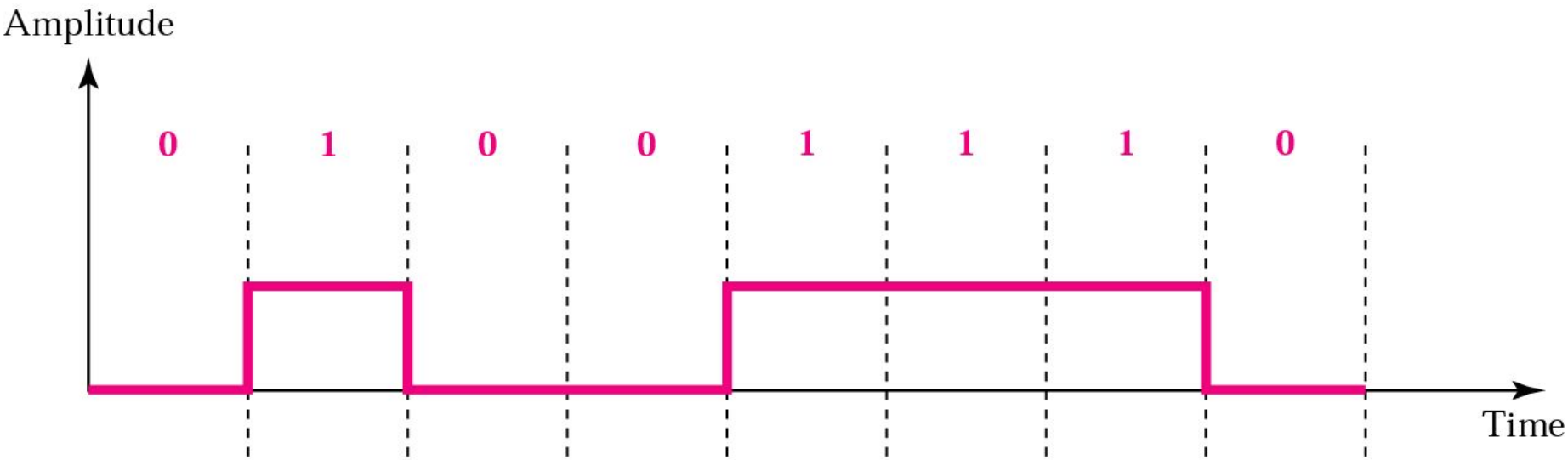


In unipolar encoding, we use only one voltage level.

- The unipolar encoding is named so because it uses only one polarity.
- Only one of the binary states is encoded, usually the 1. The other state, usually the 0, is represented zero voltage, or an idle line.

Unipolar encoding is inexpensive to implement.

Unipolar Encoding



- Unipolar encoding has 2 types of disadvantages that makes it unusable :

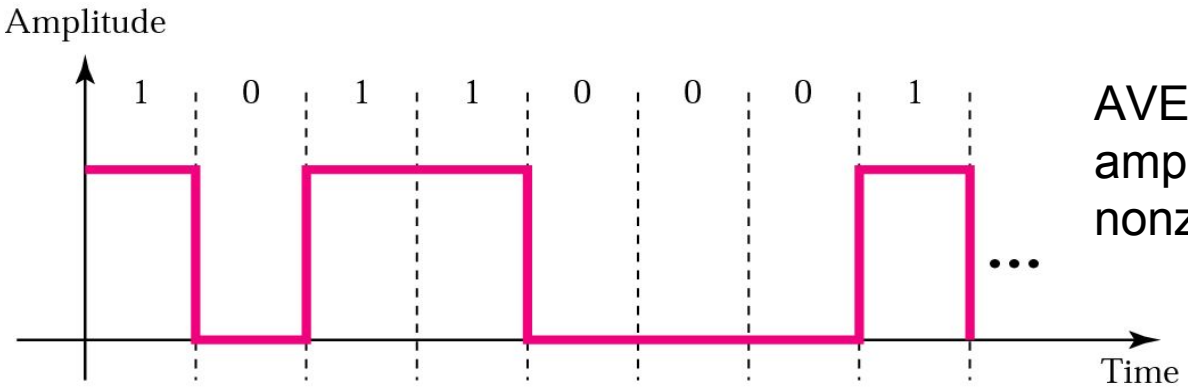
1. **DC component**

2. **Synchronization**

1. DC component

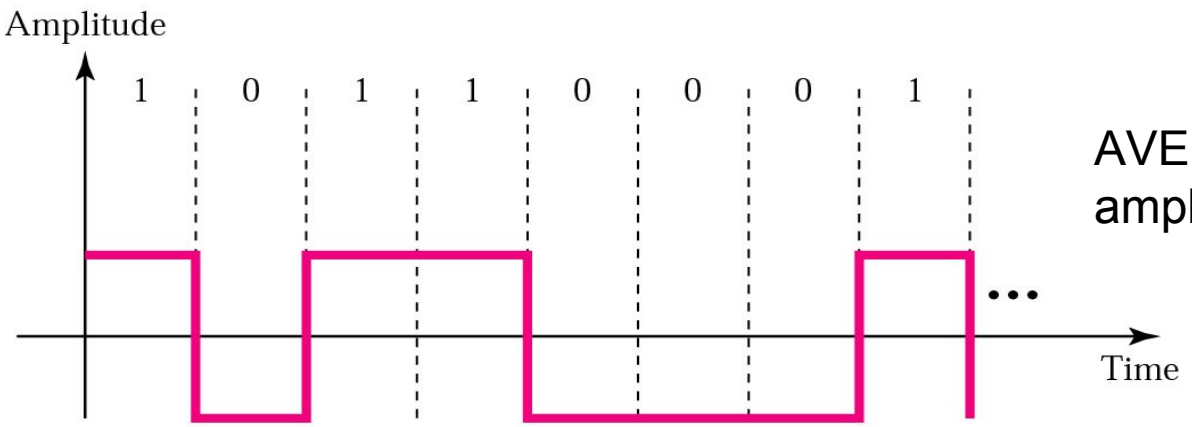
- The average amplitude of unipolar encoded signal is nonzero.
- This creates a direct current(DC) component(i.e., a component with zero frequency means no frequency) .
- But we need a frequency to send signals which is not present in DC.
- Because of this, the signal cannot travel through media that can’t handle DC components like microwaves and transformers.

DC Component



AVERAGE amplitude is nonzero

a. A signal with dc component



AVERAGE amplitude is zero.

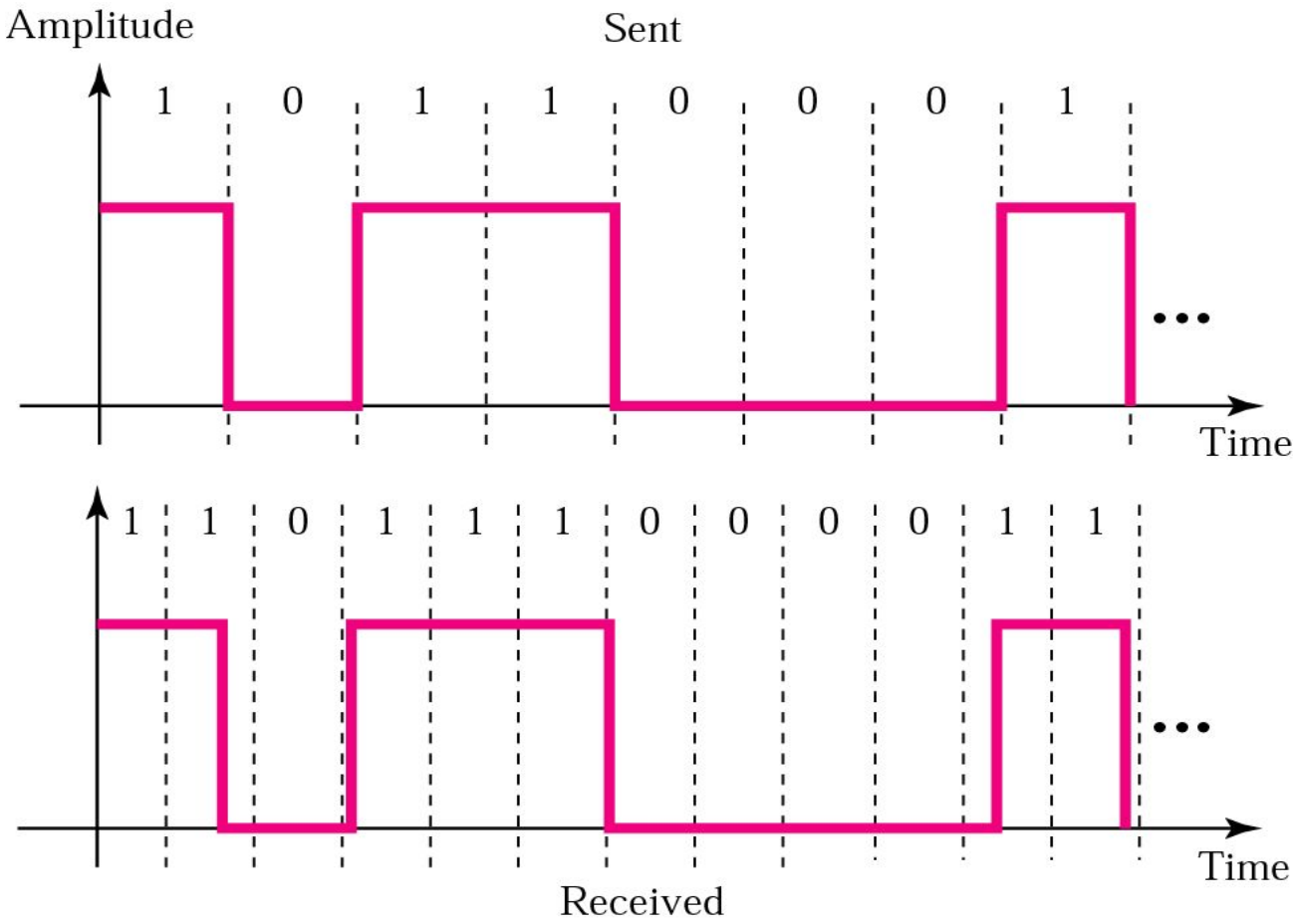
b. A signal without dc component

2. Synchronization

- When the signal is unvarying(a long uninterrupted series of 0's and 1's), receiver cannot determine the beginning and ending of each bit causing synchronization problems.

- To indicate the start of the next bit in the sequence – whenever there is no signal change the receiver has to rely on a timer.
- So this delay results in error.

Lack of Synchronization

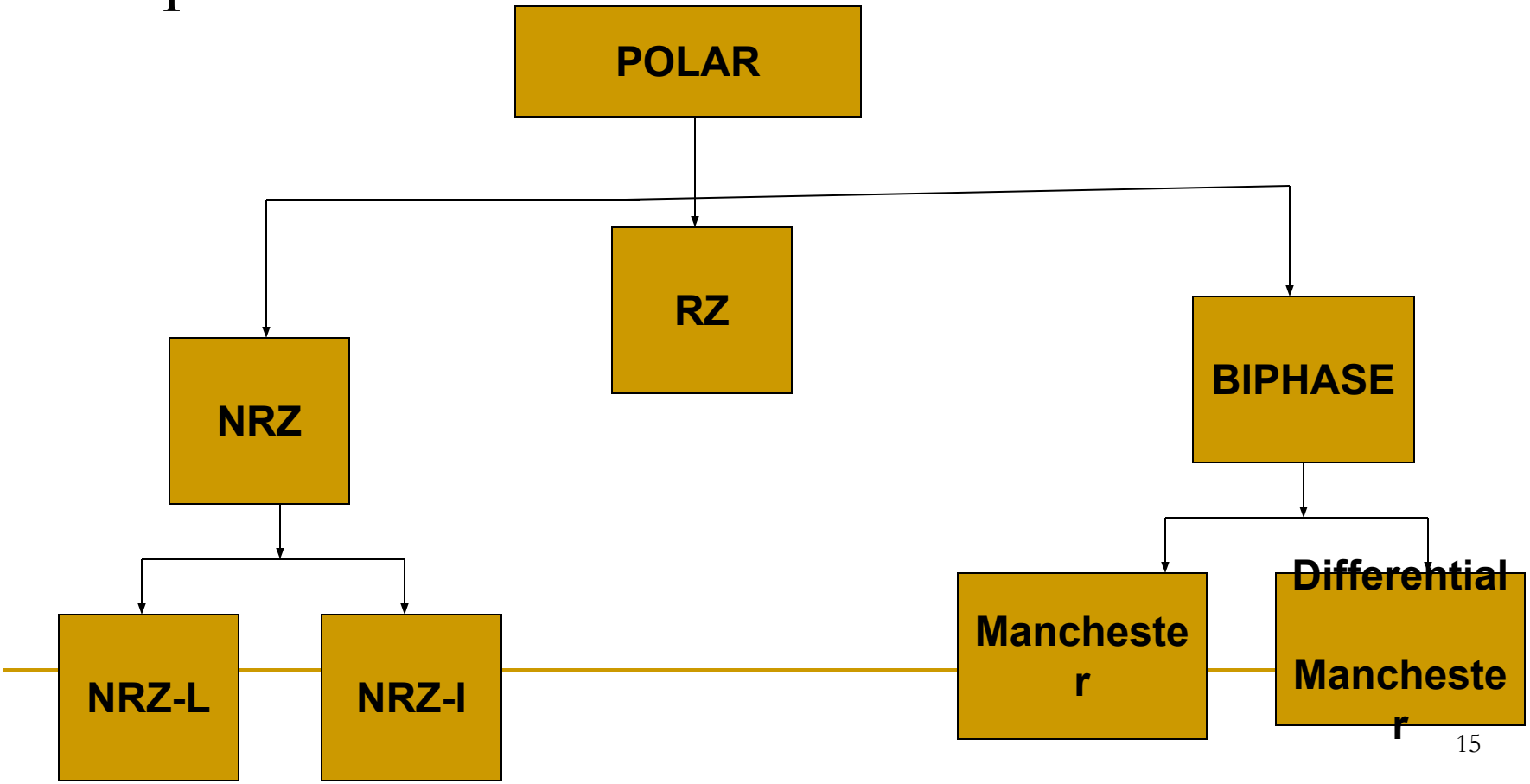


In polar encoding, we use two voltage levels: one positive & one negative

By using two level in polar encoding the DC component problem in unipolar encoding is eliminated.

Most popular POLAR encoding are :

- Non-return to zero (NRZ)
- Return to zero (RZ)
- Biphase



In NRZ, SIGNAL levels: positive & negative

- **The absence of voltage can be used to represent binary 0.**
- **Constant positive voltage used to represent binary 1.**
- **This code is known as Non Return to zero –level (NRZ-L).**
- **Unlike unipolar encoding, where 0 bit is represented by an idle line, in NRZ systems if the line is idle it means no transmission is occurring at all.**
- **In NRZ-L encoding the level of the signal determines the value of the bit.**

Non Return to zero – Level (NRZ-L)

- In NRZ-L encoding, the level of the signal determines the value of the bit.

- A positive voltage usually means the bit is a 0, while a negative voltage means the bit is a 1.

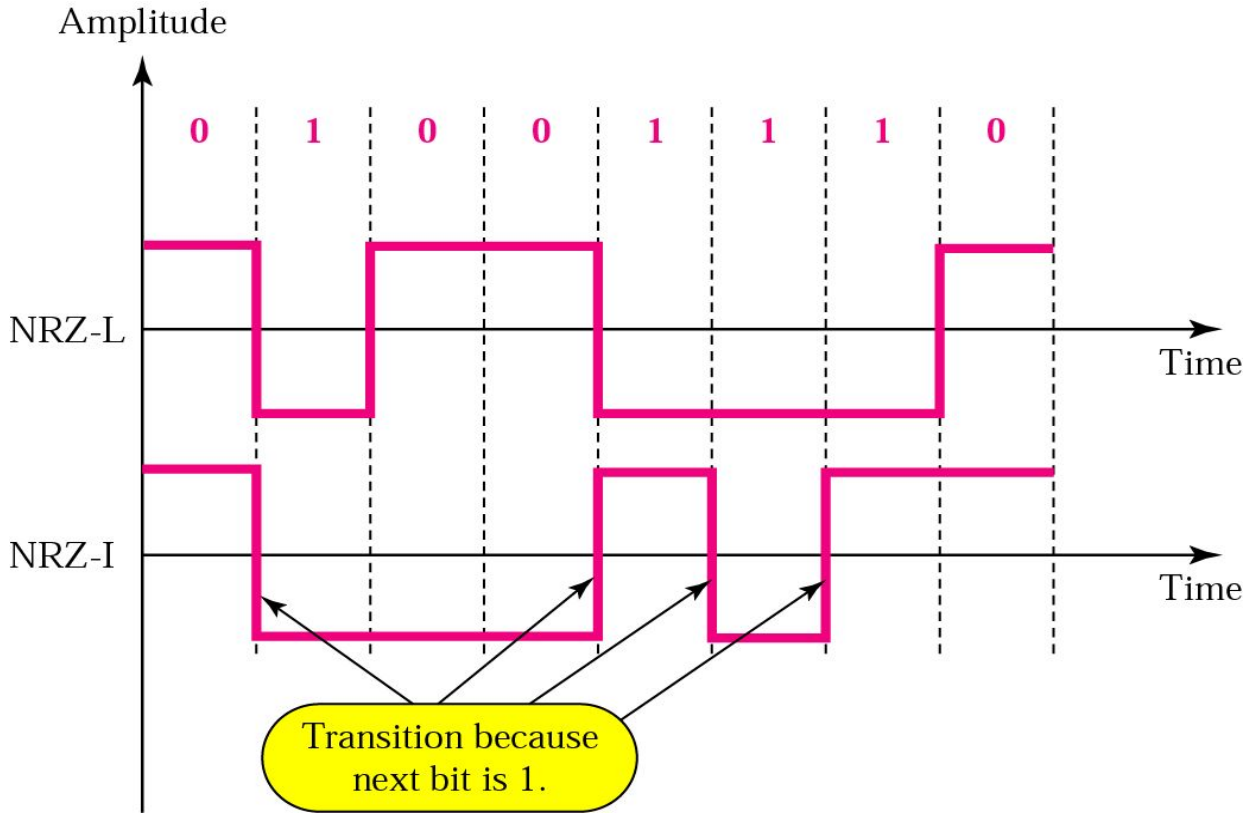
- Thus, the level of the signal is dependent upon the state of the bit.

Non Zero to zero – INVERT (NRZ-I)

In NRZ-I an inversion of the voltage level (not the level) represents a 1 bit.

- There is no positive and negative values.
- It is the transition between a positive and a negative voltage, not the voltage itself that represents a 1 bit.
- A 0 bit is represented by no change.
- An advantage of NRZ-I over NRZ-L is that because the signal changes every time a 1 bit is encountered it provides some synchronization.
- Since series of 1's cause several inversions.

Polar: NRZ-L and NRZ-I Encoding



Return to zero (RZ)

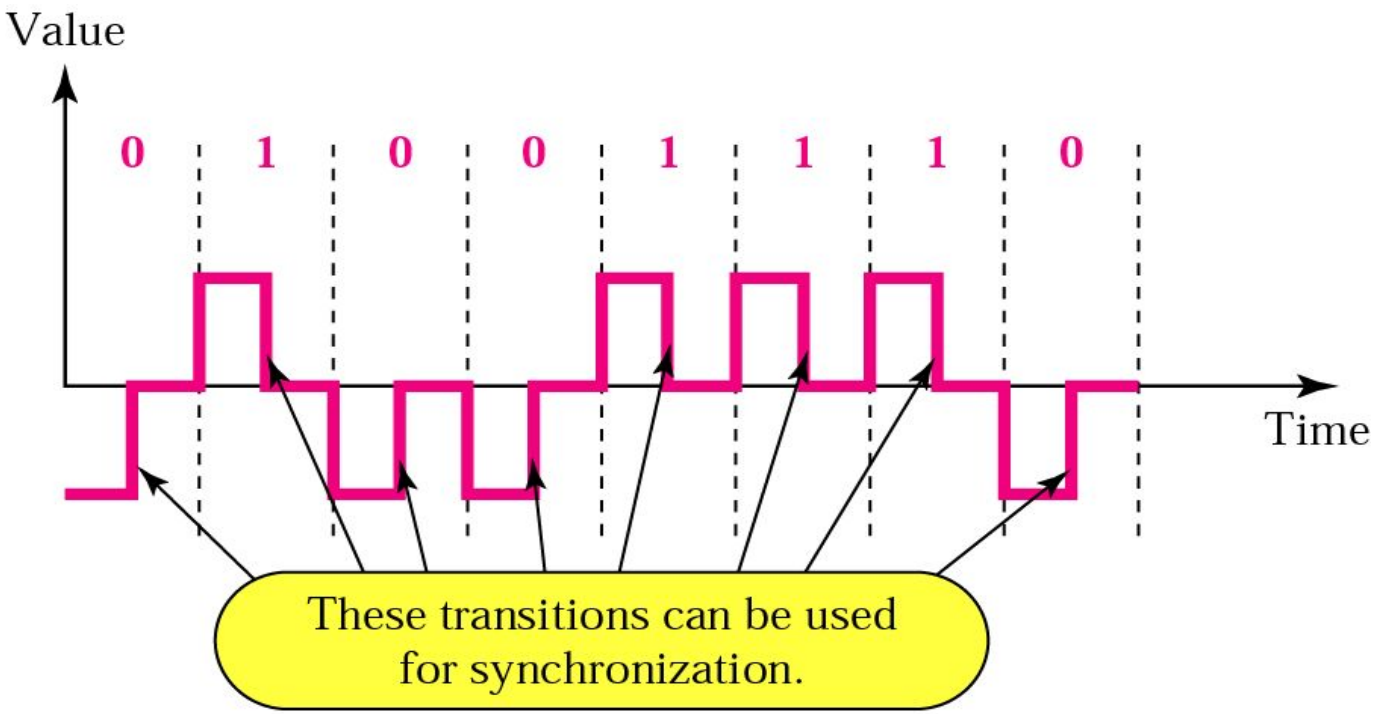
- The main problem with NRZ encoding occurs when the sender and the receiver clocks are not synchronized.
- The receiver doesn't know when one bit has ended and the next bit starting.
- Anytime the original data contain strings of consecutive 1's or 0's, the receiver can lose its place.
- To ensure synchronization, there must be a signal change for each bit.
- The receiver can use these changes to build up, update and synchronize its clock.
- Solution - Return to zero (RZ) encoding, which uses 3 values- positive , negative and zero.

- In NRZ, the signal changes not between bits but during each bit in.

- Like NRZ-L ,a positive voltage means 0 and a negative voltage means 1 returning to zero.

- A 1 bit is represented by positive – to – zero and a 0 bit by negative – to- zero, rather than by positive and negative along.

Polar: RZ Encoding



Bi –Phase Codes

Bi- phase codes –The signal changes at the middle of the bit interval but instead of returning to zero, it returns to the opposite pole.

<u>Advantages:</u>

Synchronization – with a predictable transition per bit time the receiver can "synch" on the transition [self-clocking].

No DC component

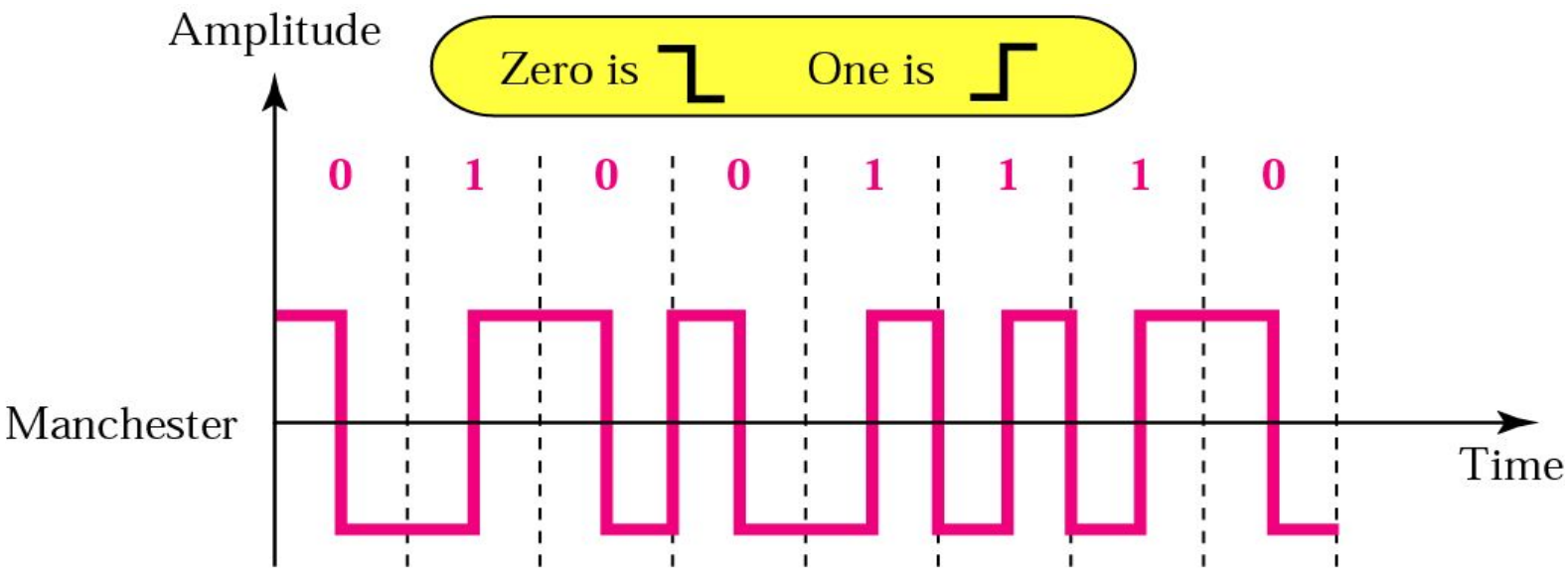
Error detection – the absence of an expected transition can used to detect errors.

In Manchester Encoding,

- The duration of the bit is divided into two halves.
- The voltage remains at one level during the first half and moves to the other level during the second half.
- the transition at the middle of the bit is used for both synchronization and bit representation.

- A negative to positive represents binary 1, and a positive to negative transition represents binary 0.

Polar: Manchester Encoding



Differential Manchester encoding

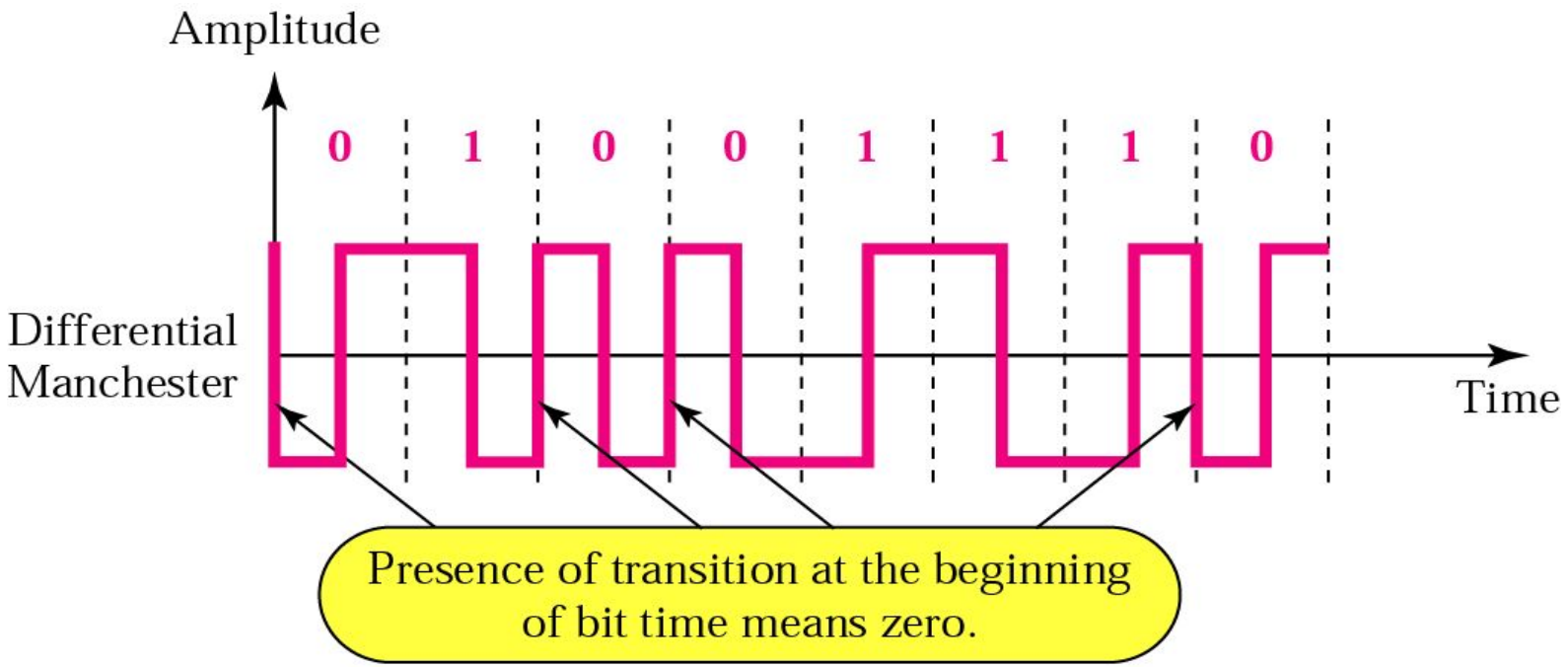
Encoding in which (a) dataEncoding in which (a) data and clockEncoding in which (a) data and clock signals are combined to form a single self-synchronizingEncoding in which (a) data and clock signals are combined to form a single self-synchronizing data stream,

In Differential Manchester Encoding, the transition at the middle of the bit is used only for synchronization.

The bit representation is defined at the beginning of the bit.

Transition means Binary 0 and No transition means Binary 1.

Polar: Differential Manchester Encoding



BIPOLAR

In bipolar encoding, we use three levels: positive, zero, and negative.

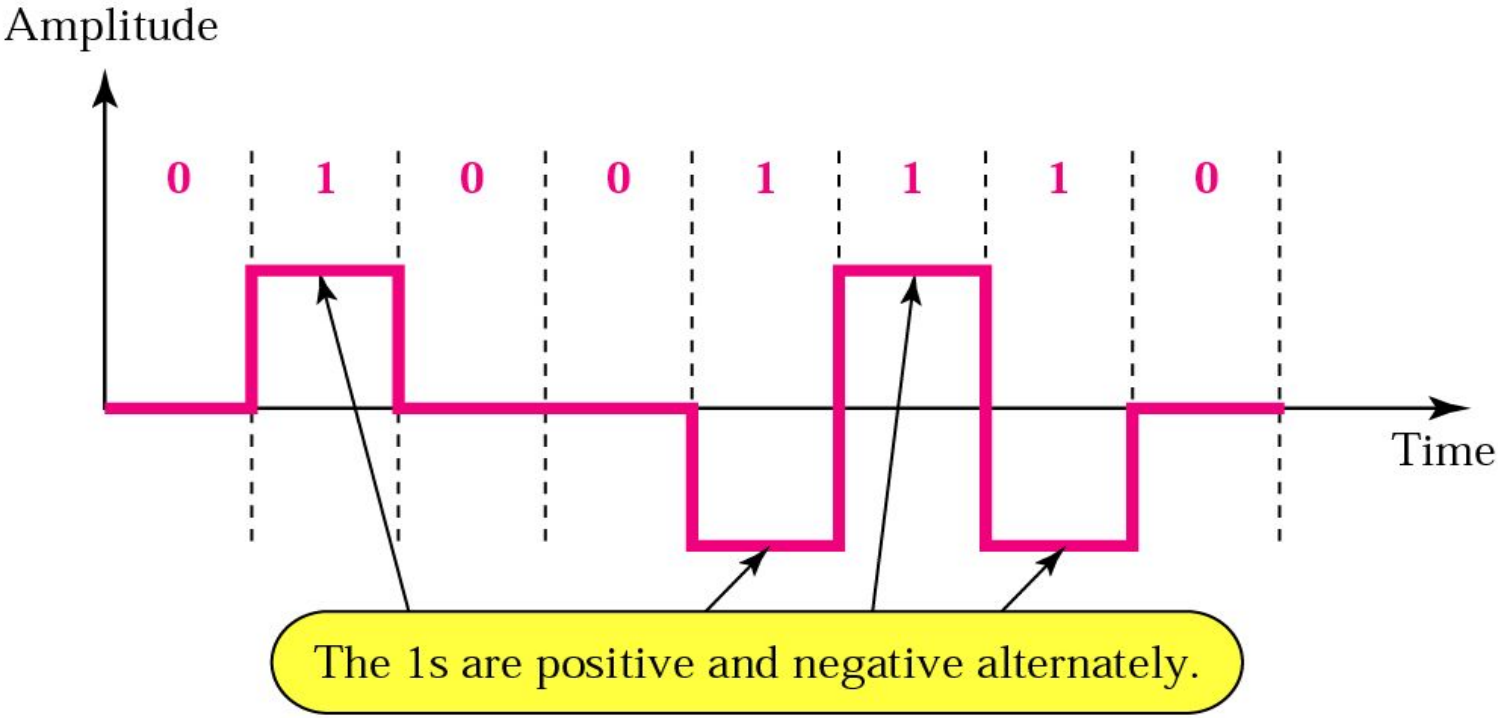
- The zero level in bipolar encoding is used to represent binary 0.

- The 1's are represented by alternating positive and negative voltages.

- If the first 1 bit is represented by the positive amplitude, the second will be represented by the negative amplitude and so on.
- The common bipolar encoding scheme are AMI, BnZS and HDB3.

- It is the simplest type of bipolar encoding.
- AMI means alternative 1 inversion.
- A zero represents binary 0.
- Binary 1 is represented by alternating positive and negative voltages.
- AMI fails to achieve synchronization.

Bipolar: AMI (Alternative Mark Inversion) Encoding



Bipolar n-Zero Substitution (BnZS)

- To solve the problem of synchronization – for long distance transmission.
- In this scheme, whenever n consecutive zeros occur in the sequence, some of the bits in these n bits become positive or negative which helps synchronization.

Example:-

BnZS is B8ZS.

- In this technique, eight consecutive zero-level voltages are replaced by the sequence – 000VB0VB.

- The V in the sequence denotes violation; this is a nonzero voltage that breaks an AMI rule of encoding

- The B in the sequence denotes bipolar; which means a nonzero level voltage in accordance with the AMI rule.

ANALOG TO DIGITAL ENCODING

1. **Pulse Amplitude Modulation (PAM)**

2. **Pulse Code Modulation (PCM)**

1. Pulse Amplitude Modulation (PAM)

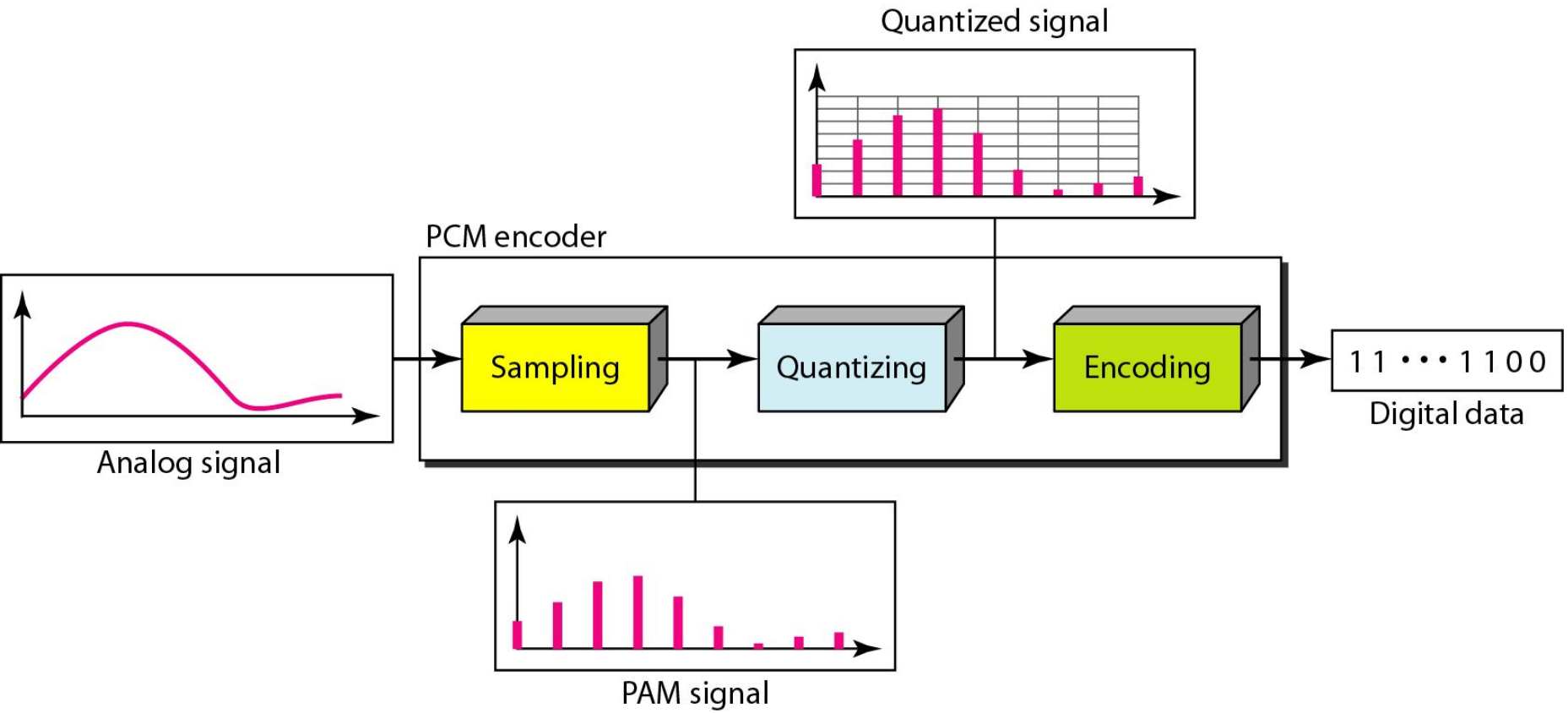
- The first step in analog- to- digital conversion.
- This technique takes an analog signals, **samples** it and generates a series of pulses based on the results of the sampling.
- **Sampling means measuring the amplitude of the signal at equal interval.**
- **PAM technique – <u>sample and hold.</u>**
- Means at a given movement, the signal level is read, and then held briefly.
- PAM is not useful to data communication because even though it translates the original waveform to a series of pulses, these pulses are still of any amplitude
- To make them digital, we must modify them by using PCM.

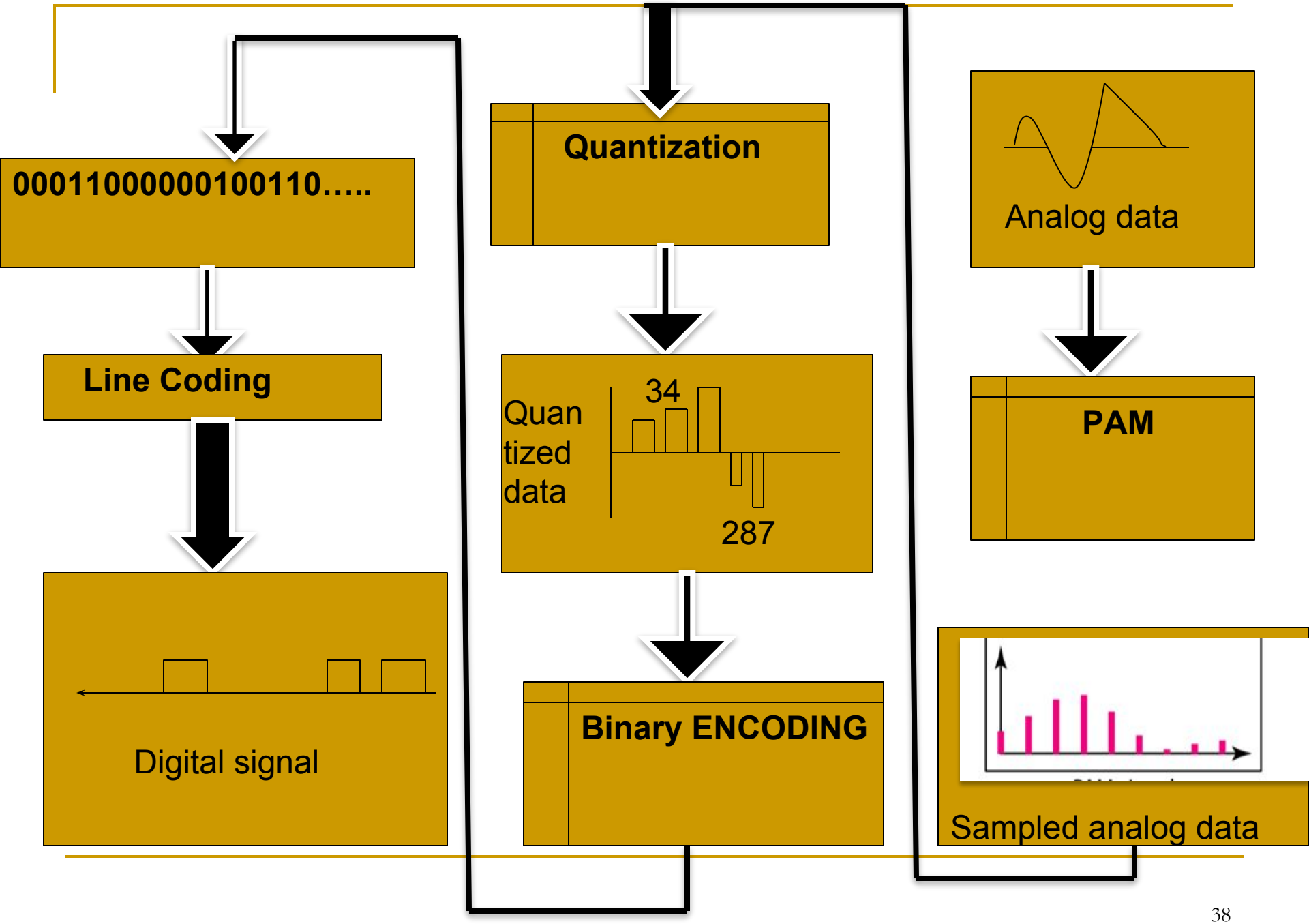
2. Pulse Code Modulation (PCM)

- PCM modifies the pulses created by PAM to create a completely digital signal.
- Steps : -
  - The analog signal is sampled.
  - The sampled signal is quantized.
  - **Quantization is a method of assigning integral value in a specific range to sampled instances.**
  - Each quantized values are translated into seven-bit binary equivalent. The eight bit represents the sign.
  - The binary digits are then transformed into digital signals using one of the digital – to – digital techniques.
  - PCM actually made up of four separate processes :

    1) PAM 2) Quantization 3) Binary encoding 4) Line coding

PCM


Quantized signal
PCM encoder
Sampling
Quantizing
Encoding
1 1 • • • 1 1 0 0
Digital data
Analog signal
PAM signal


00011000000100110…..
Line Coding
Digital signal
Quantization
Quan
tized
data
34
287
Binary ENCODING
Analog data
PAM
Sampled analog data

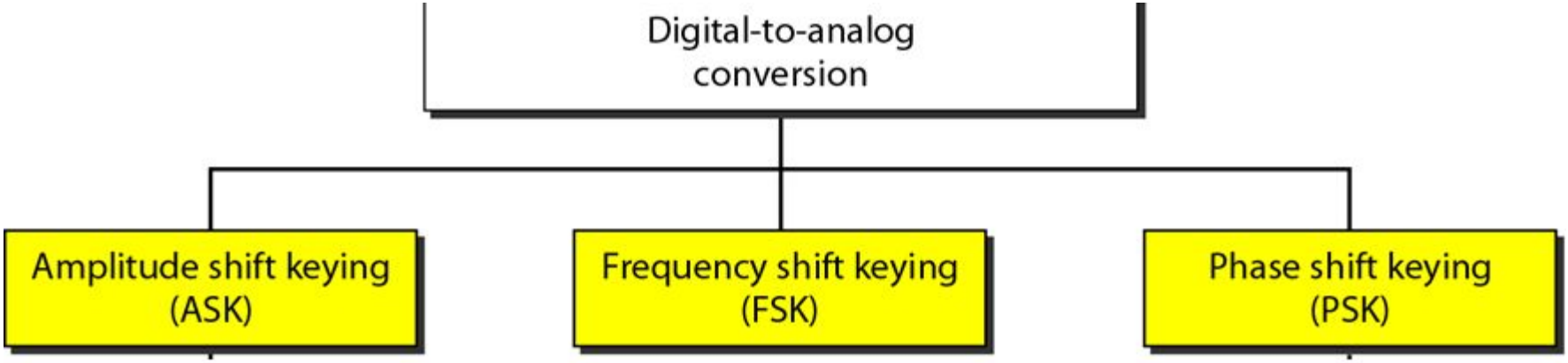
DIGITAL – TO – ANALOG ENCODING

- Is the process of changing one of the characteristics of an analog signal based on the information in a digital (0’s and 1’s).

DIGITAL – TO – ANALOG ENCODING

Three mechanisms for encoding digital data into an analog signals are :

1) Amplitude Shift Keying (ASK)
2) Frequency Shift Keying (FSK)
3) Phase Shift Keying (PSK)


Digital-to-analog conversion
Amplitude shift keying (ASK)
Frequency shift keying (FSK)
Phase shift keying (PSK)

Bit rate, N, is the number of bits per second (bps).
Baud rate is the number of signal
elements per second (bauds).

In the analog transmission of digital data, the signal or baud rate is less than or equal to the bit rate.

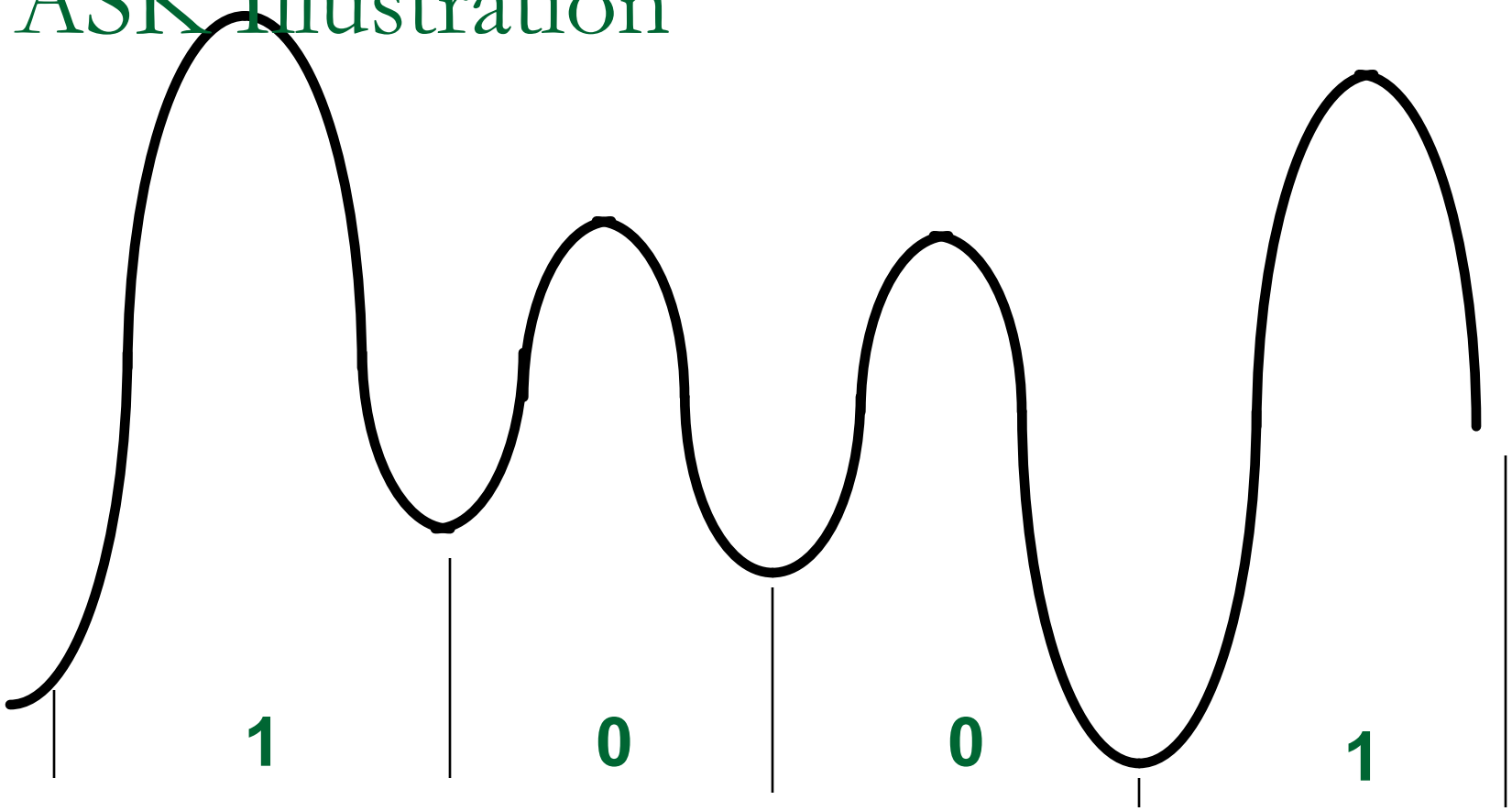
Amplitude Shift Keying (ASK)

- Both frequency and phase remain constant while the amplitude changes.
- The voltage level to represent binary 1 and 0 is left to the system designers.
  For example: a digital “1” could not affect the signal, whereas a digital “0” would, by making it zero.
  The line encoding will determine the values of the analog waveform to reflect the digital data being carried.

- The binary is represented by low voltage/ amplitude and binary 1 is represented by a high voltage.

- The noise interference is more.

ASK Illustration

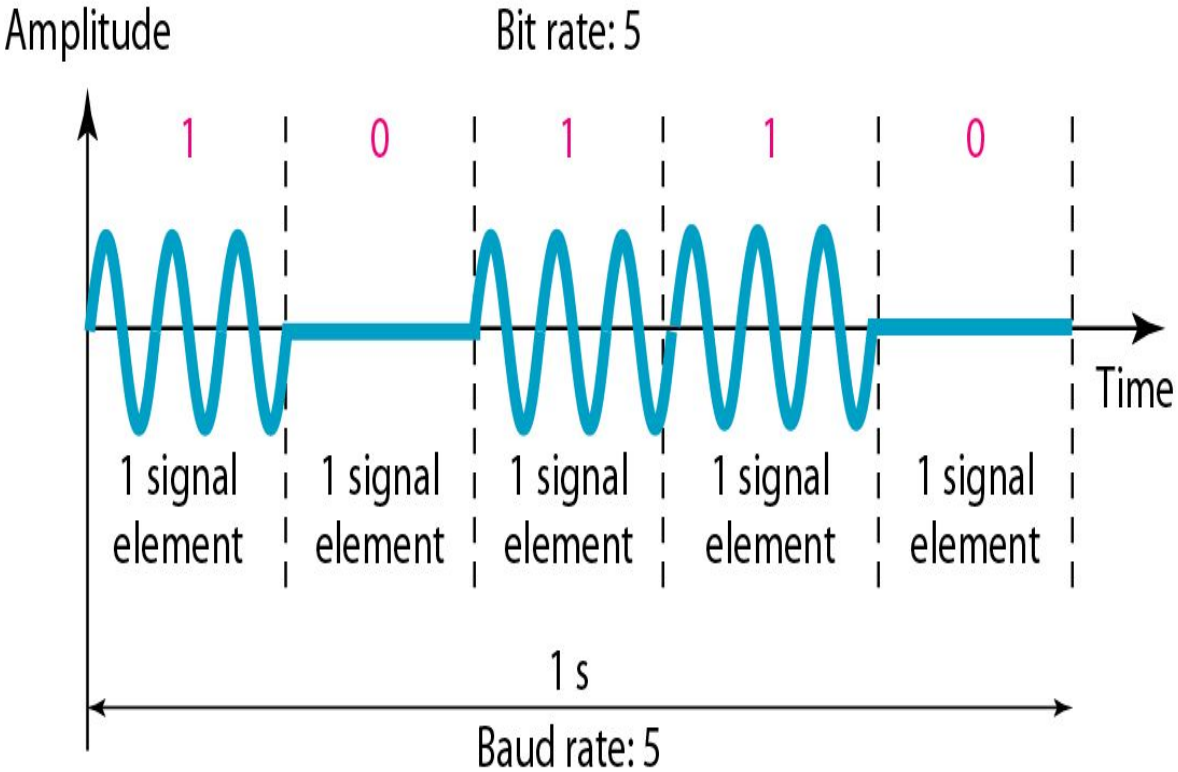


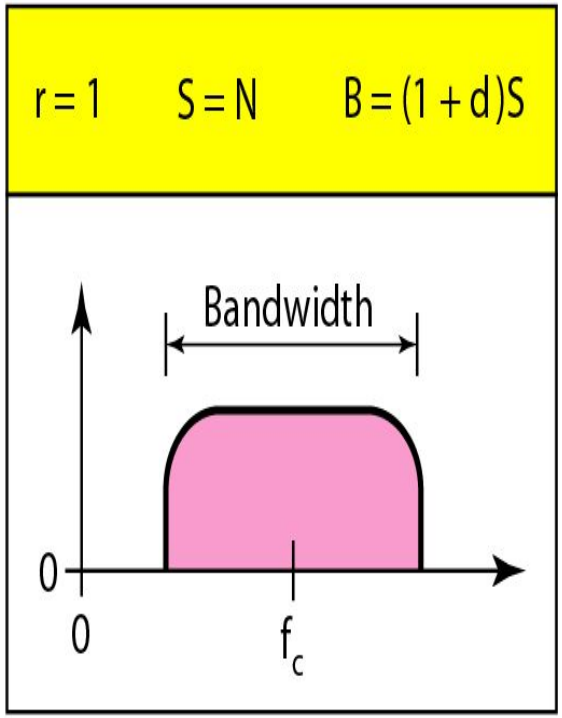
Bandwidth of ASK

- The bandwidth B of ASK is proportional to the signal rate S.

$$B = (1+d)S$$

- “d” is due to modulation and filtering, lies between 0 and 1.


Amplitude
Bit rate: 5
1
0
1
1
0
Time
1 signal element
1 signal element
1 signal element
1 signal element
1 signal element
1 s
Baud rate: 5

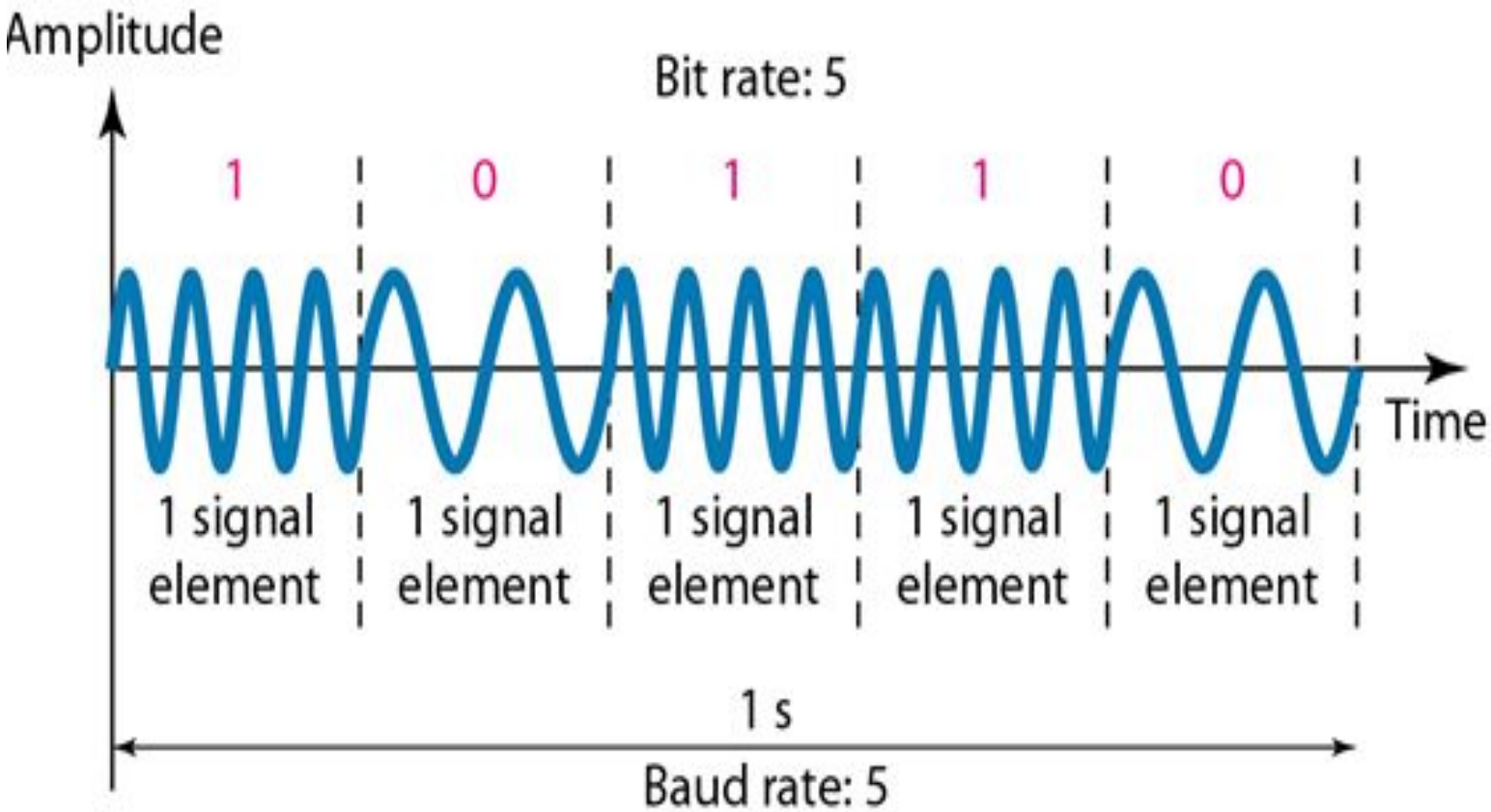

r = 1
S = N
B = (1 + d)S
Bandwidth
0
0
f_c

Frequency Shift Keying

- In FSK the frequency of the carrier signal is varied to represent binary 1 0r 0. f_c

- Both amplitude and phase remains constant.

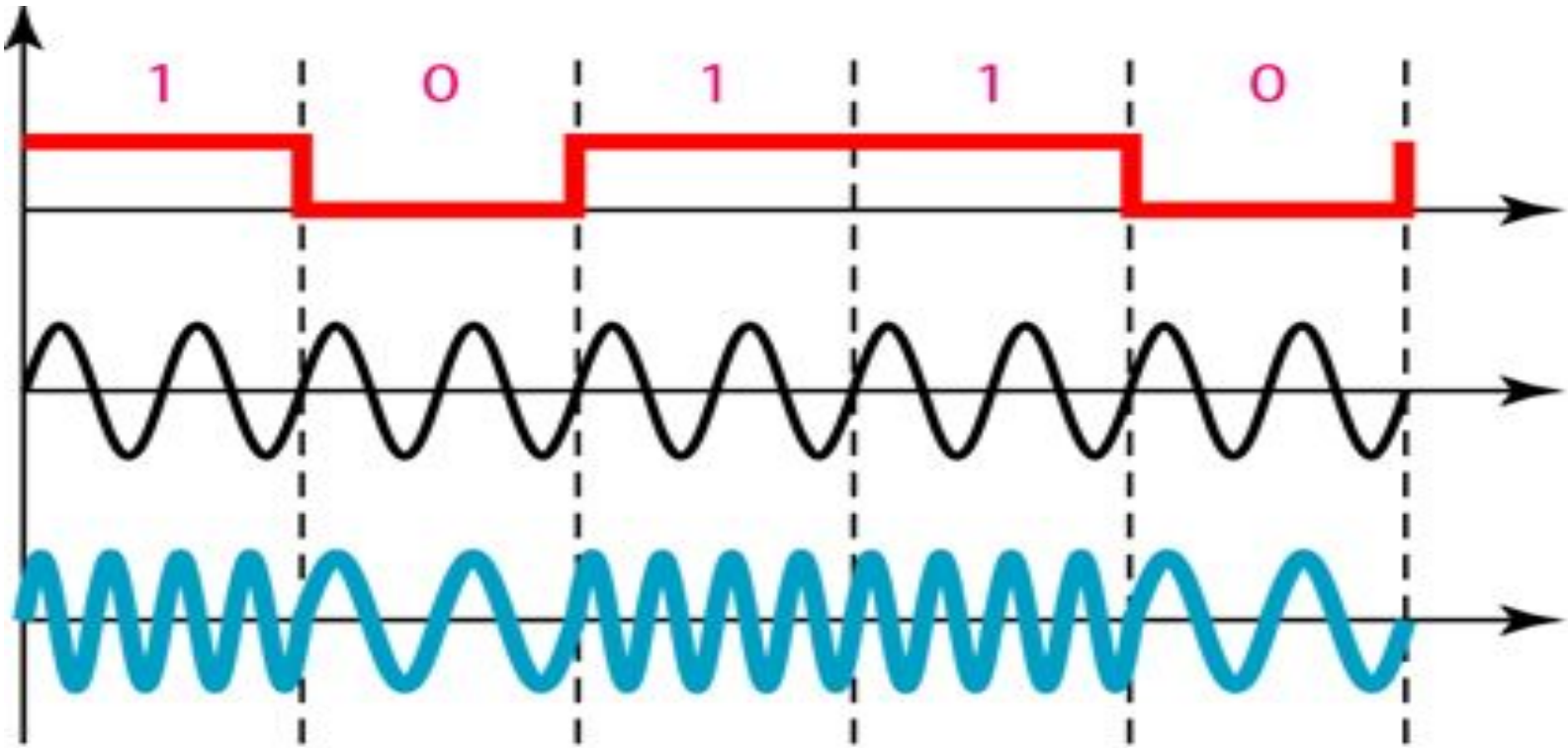
- FSK avoids most of the problems from noise.


Amplitude
Bit rate: 5
1
0
1
1
0
Time
1 signal element
1 signal element
1 signal element
1 signal element
1 signal element
1 s
Baud rate: 5

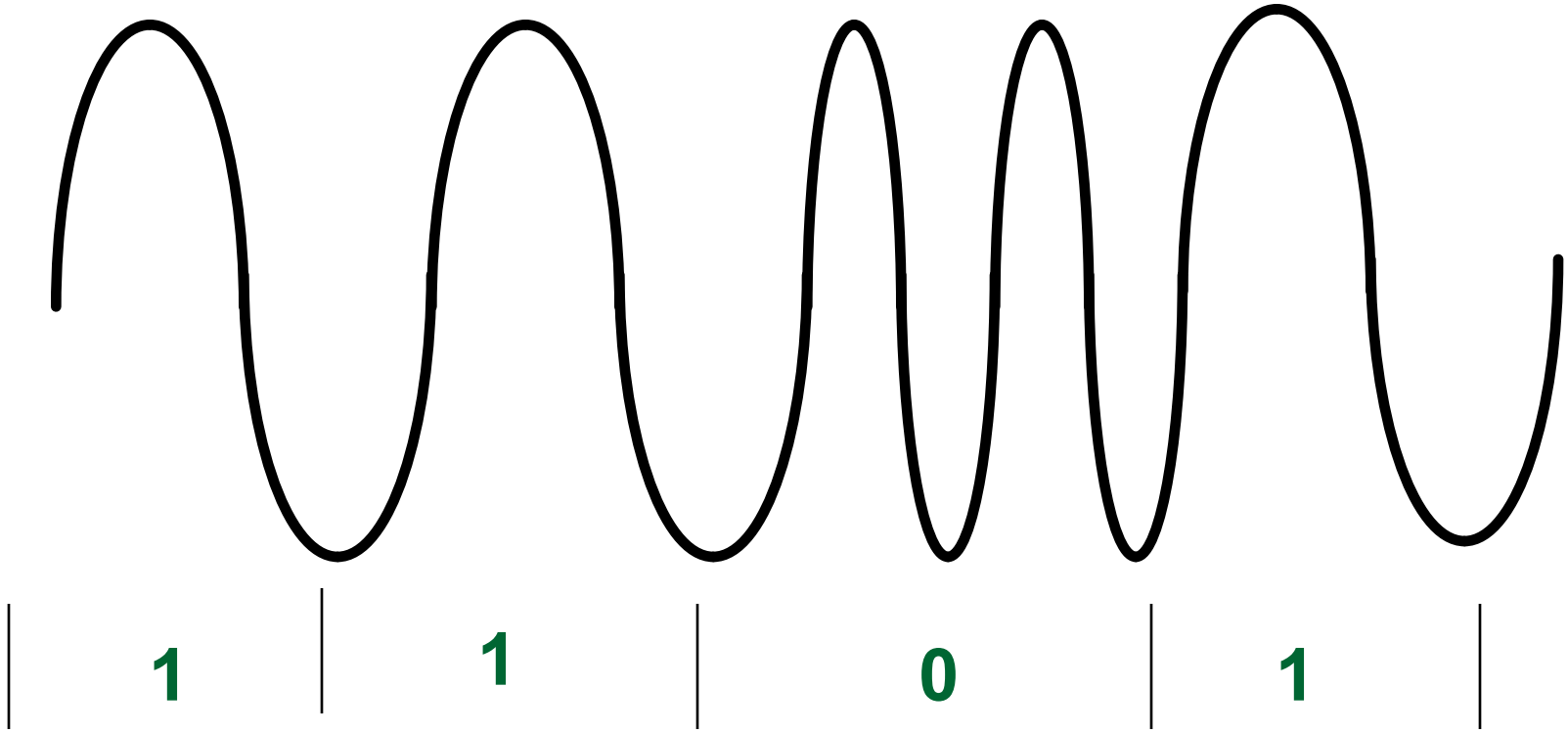
Bandwidth of FSK

- If the difference between the two frequencies (f_1 and f_2) is 2f, then the required BW B will be:

$$B = (1+d)xS + 2f$$


1
0
1
1
0

FSK Illustration

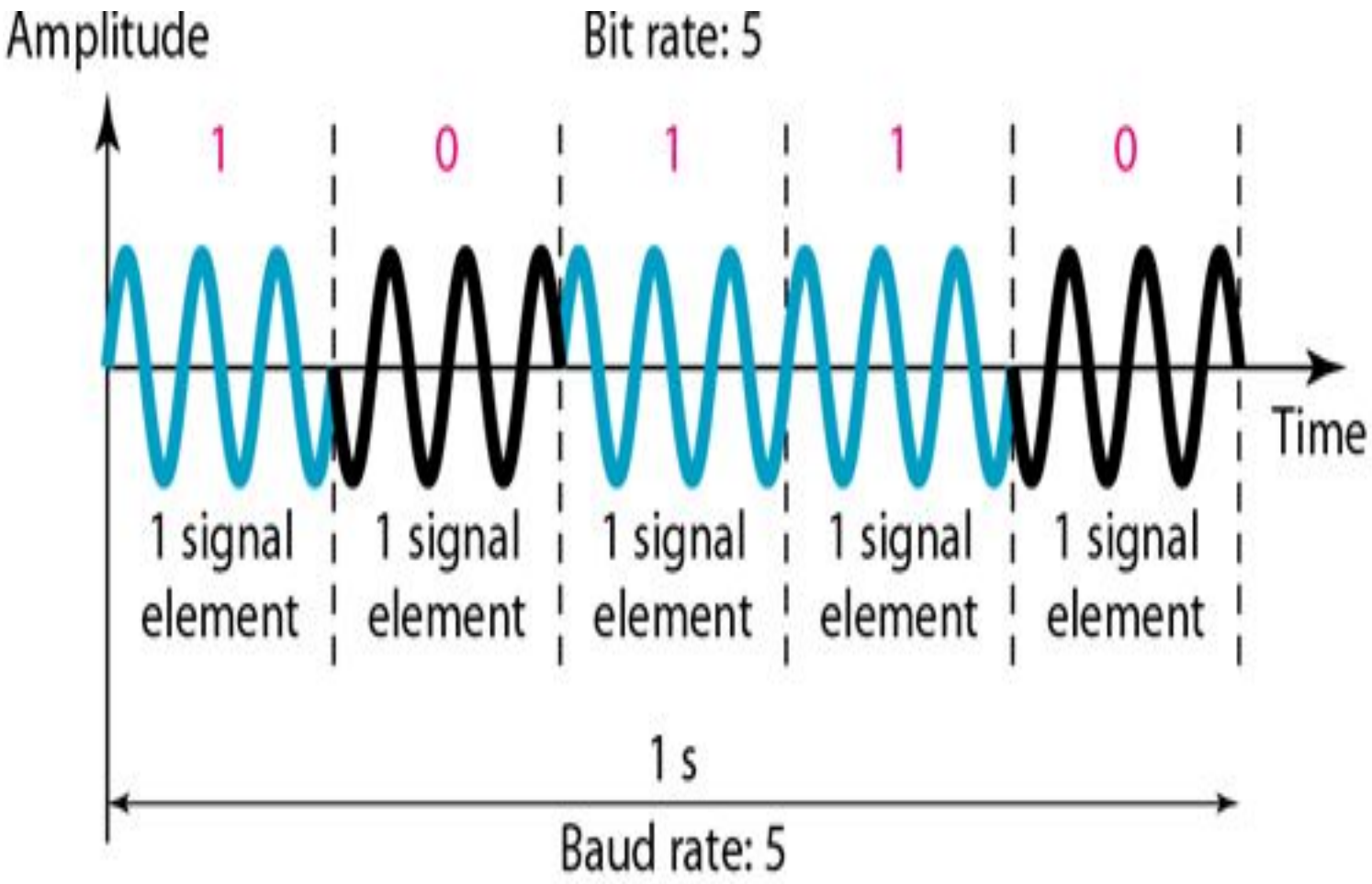


Phase Shift Keying

- The phase shift of the carrier signal to represent digital data to represent binary 1 or 0.
- Both amplitude and frequency remain constants.
- Example, if we start with a phase of 0 degrees to represent binary 0, then we can change the phase to 180 degrees to send binary 1.
- The bandwidth requirement, B is:

$$B = (1+d)xS$$

- PSK is much more robust than ASK as it is not that vulnerable to noise.


Amplitude
Bit rate: 5
1
0
1
1
0
Time
1 signal element
1 signal element
1 signal element
1 signal element
1 signal element
1 s
Baud rate: 5

ANALOG – TO- ANALOG ENCODING

Many times it is necessary to modulate analog data onto a different set of analog frequencies.

- Broadcast radio and television are two very common examples of this.

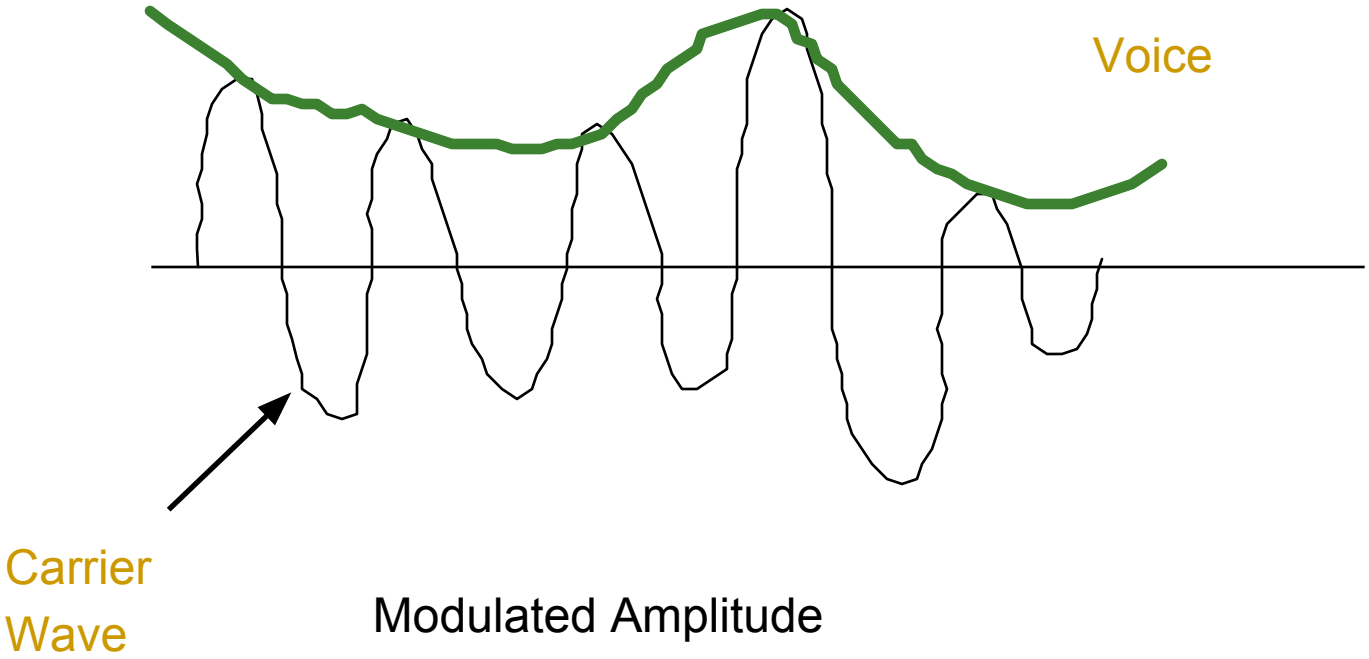
Three ways :

1) Amplitude modulation (AM)
2) Frequency modulation (FM)
3) Phase modulation (PM)

Amplitude modulation (AM)

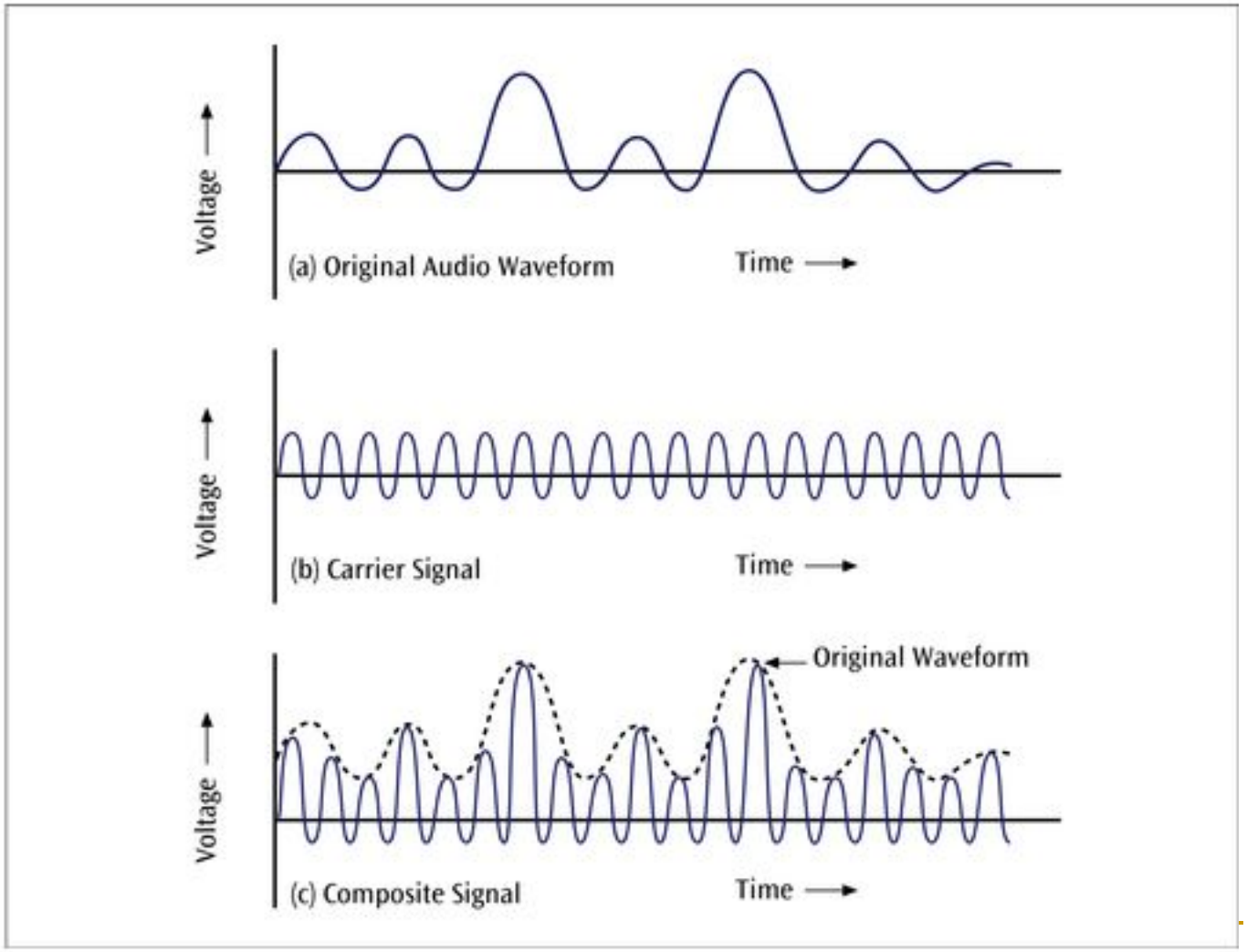
- In AM transmission, only the amplitude of the carrier signal is modulated to contain the information to be transmitted.

- The frequency and phase of the carrier remain the same, only the amplitude changes to follow variations in the information.



Modulated Amplitude

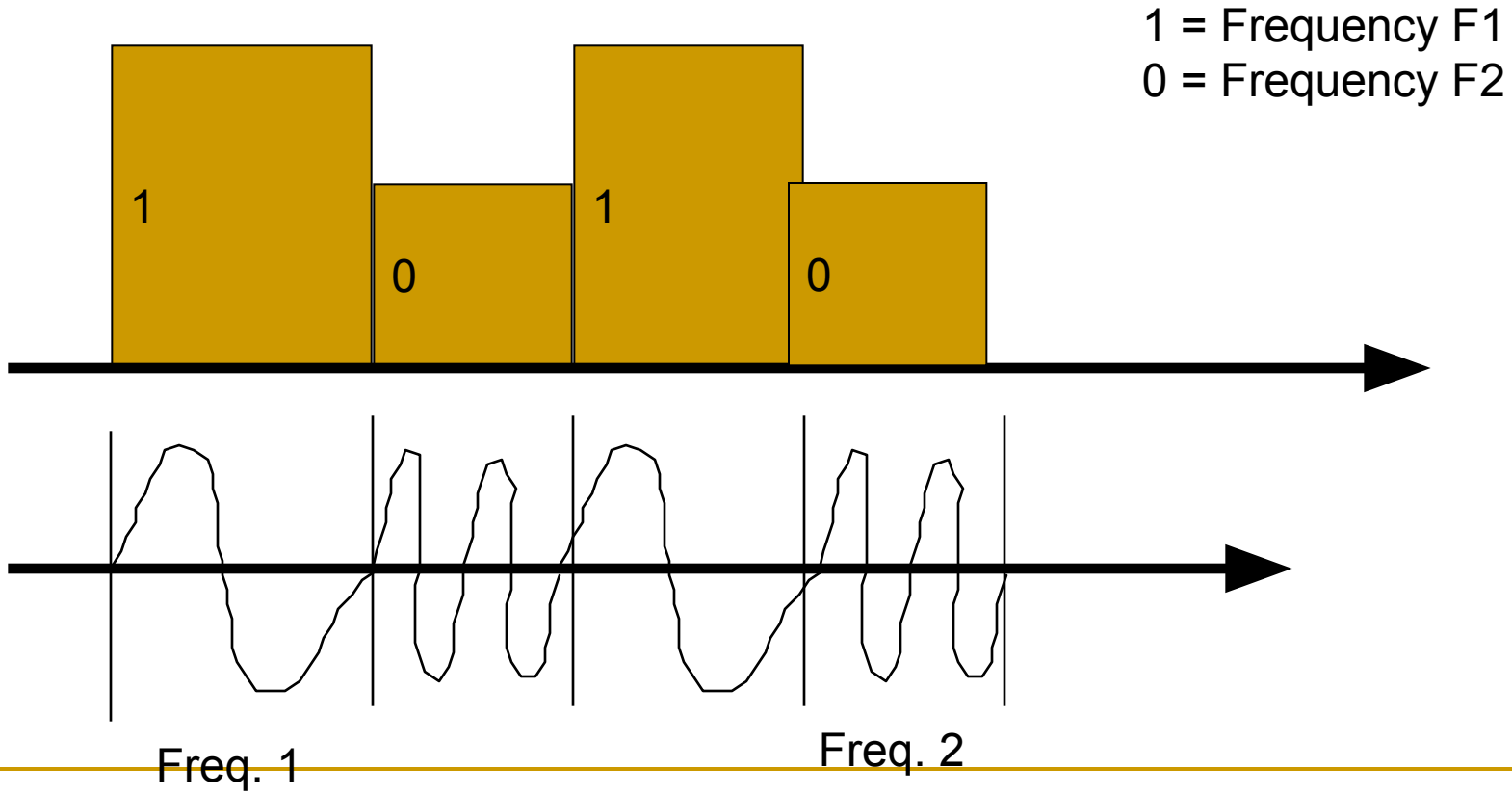
Figure 2-26
An audio waveform modulated onto a carrier frequency using amplitude modulation



Frequency modulation (FM)

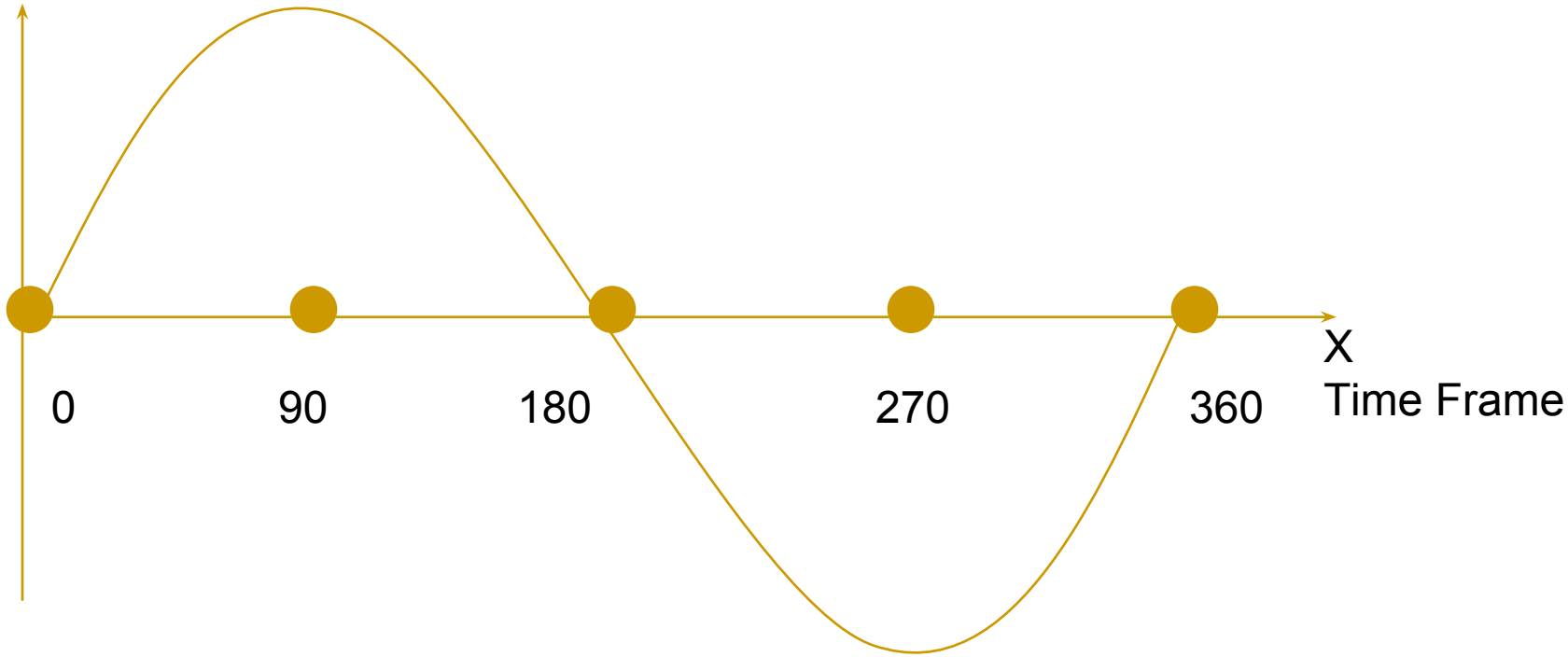
- Frequency is modulated
- Frequency f1
  - Represents 1
- Frequency f2
  - Represents 0
- The amplitude and phase remains unaltered in both cases.
- A variation of the FM technique described here is used in FM radio transmission

Frequency Modulation (FM)



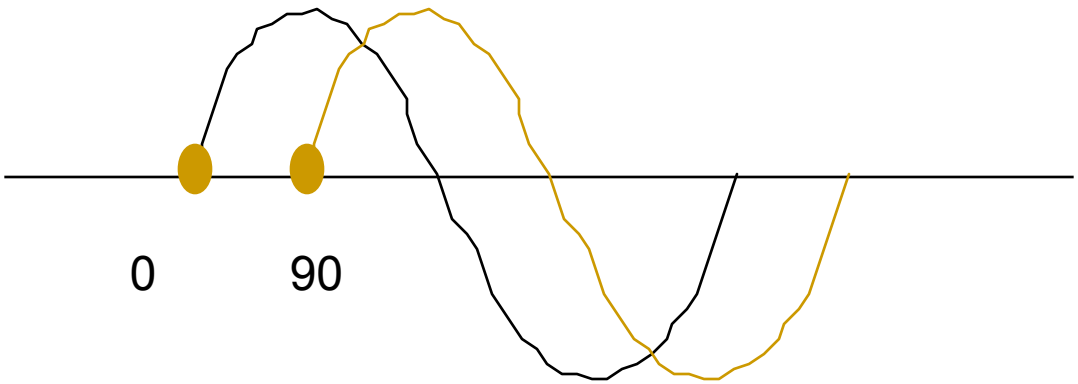
Phase modulation (PM)

- The phase is modulated.
- Amplitude and frequency of the carrier signal remain constant.
- Phase shift of 0 represents a 0
- Phase shift of 90 degrees represents a 1
- Also known as Phase Shift Keying, it is used in a number of modern modems as well

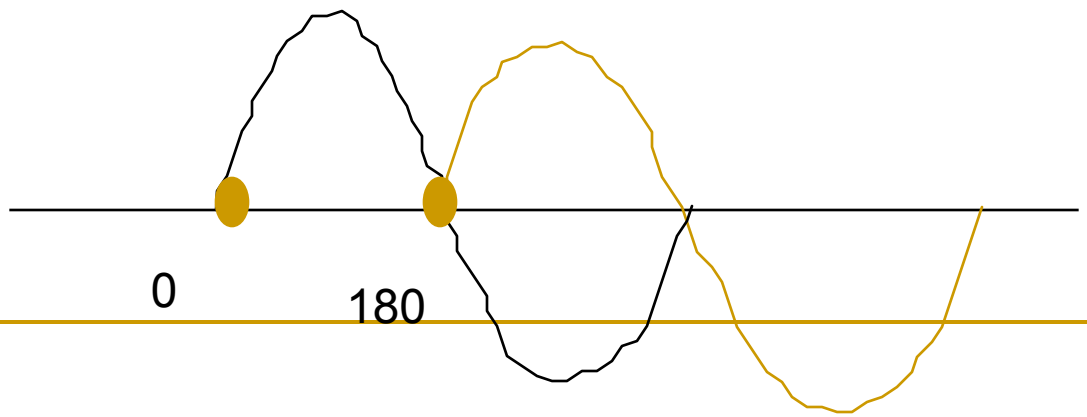

0
90
180
270
360
X
Time Frame

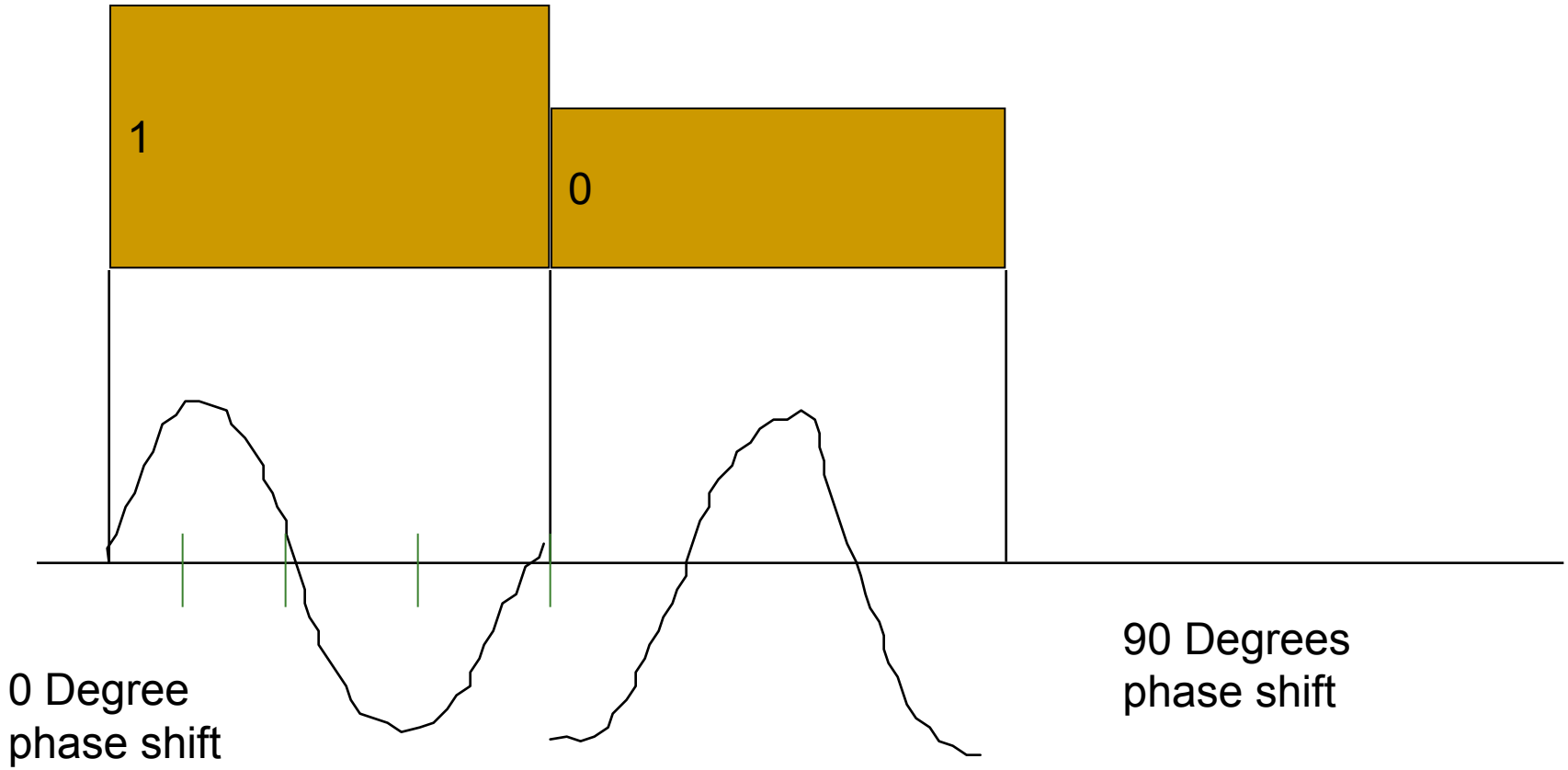
The Concept of Phase Shift

90 degrees phase shift



180 degrees phase shift




1
0
0 Degree
phase shift
90 Degrees
phase shift